IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2023

MATHEMATICS 1A

Wednesday 11 January 2023 10:00 am

Time allowed: 2:00 hours

This exam has two sections.

Section A is worth 40 Marks, and is multiple choice, please enter your answers on the multiple-choice answer sheet. Please note that some of the questions are worth 2 marks, and some 3 marks.

Section B has three longer questions, worth 20 marks each. Enter your answers in the booklet provided.

No calculators allowed. Formula sheet provided

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ELEC40010 - MATHEMATICS1A

Section A: Multiple Choice

Enter your answers on the multiple-choice answer sheet. Please note that some of the questions are worth 2 marks, and some 3 marks. Total marks for this section: 40.

1. In complex exponential form, $re^{i\theta}$, the complex number $\dfrac{i-1}{i+1}$ has

   (a) $r = 1/2$ and $\theta = \pi$, (b) $r = 1/2$ and $\theta = \pi/2$,

   (c) $r = 1$ and $\theta = -\pi/2$, (d) $r = 1$ and $\theta = \pi/2$,

   (e) none of the above [3]

2. The roots of the equation $z^3 + z^2 + z + 1 = 0$ are

   (a) one real root, and two complex conjugate roots ,

   (b) two real roots, and one complex root ,

   (c) unobtainable , (d) three real roots ,

   (e) none of the above [3]

3. Three functions are given by

$$(I)\ x\cos\left(\frac{\pi}{2} - x\right) , \quad (II)\ \sin[\ln(1+x^2)] , \quad (III)\ \frac{\cos x}{x} .$$

   Assuming the domains are defined for each function, which of the following statements is **true**?

   (a) (II) is not periodic and (I) is even , (b) (II) is periodic and (III) is odd ,

   (c) (III) is periodic and (II) is not , (d) (I) and (III) are both even ,

   (e) none of the above. [3]

4. The limit: $\lim_{x\to 0} \dfrac{x^2}{1 - x - e^{-x}}$, exists and is equal to

   (a) 1 , (b) -2 , (c) 3 , (d) -4 ,

   (e) none of the above. [3]

5. The function $y = \dfrac{x^2 - x + 1}{x - 1}$ has

(a) a minimum at $x = 0$, (b) a maximum at $x = 2$,

(c) a point of inflection in the interval $[2, \infty)$, (d) a diagonal asymptote ,

(e) none of the above. [3]

6. Given $x + y - \ln(x - y) = 42$, which of the following is the derivative $\dfrac{dy}{dx}$?

(a) $\dfrac{x + y - 1}{x - y + 1}$, (b) $-\dfrac{y - x + 1}{x - y - 1}$, (c) $-\dfrac{y - x + 1}{y - x - 1}$, (d) $\dfrac{x - y - 1}{x + y + 1}$,

(e) none of the above. [3]

7. The integral $I = \displaystyle\int \frac{1}{1 + \cos x}\, dx$ is equal to

(a) $\tan(x/2) + \mathrm{C}$, (b) $\cot x - \csc x + \mathrm{C}$, (c) $\dfrac{x}{1 - \sin x} + \mathrm{C}$, (d) $-\cot(x/2) + \mathrm{C}$,

(e) none of the above [3]

8. The integral $\displaystyle\int \frac{x(x-1)}{(x^2+1)(x+1)}\, dx$ is equal to

(a) $\ln(x^2 + 1) - \ln(x + 1) + \mathrm{C}$, (b) $\ln(x + 1) - \tan^{-1}(x^2 + 1) + \mathrm{C}$,

(c) $\dfrac{2}{(x+1)^2} - \ln(x + 1) + \mathrm{C}$, (d) $\ln(x + 1) - \tan^{-1} x + \mathrm{C}$,

(e) none of the above [3]

9. The power series

$$\sum_{n=1}^{\infty} \frac{(x-1)^n}{\sqrt[3]{n}}$$

has radius of convergence R and interval of convergence I, where

(a) $R = 1$ and $I = (0, 2)$, (b) $R = 0$ and $I = [1, 2)$,

(c) $R = 1$ and $I = (0, 2]$, (d) $R = 2$ and $I = [-1, 0)$,

(e) none of the above [3]

10. Given the linear first-order ordinary differential equation (ODE)

$$t\frac{dx}{dt} + 2x = t^2,$$

which of the following is an integrating factor?

(a) x^2, (b) $\frac{1}{t}$, (c) t^2, (d) e^{2t},

(e) none of the above. [2]

11. The solution of the linear ODE

$$\frac{dy}{dx} + \frac{2}{x}y = x$$

is given by

(a) $y = \frac{1}{2}x^2 + C$, (b) $x = \frac{C}{t^2} - \frac{1}{4}t^2$, (c) $y = x^2\left(x + \frac{C}{x}\right)$, (d) $y = \frac{x^2}{4} + \frac{C}{x^2}$,

(e) none of the above. [2]

12. Given the linear second-order ODE

$$4\frac{d^2x}{dt^2} - 4\frac{dx}{dt} + x = 25\sin t,$$

which of the following is the complementary function?

(a) $x_c = c_1 te^{\frac{1}{2}t} + c_2 e^{\frac{1}{2}t}$, (b) $x_c = c_1 e^{-\frac{1}{2}t} + c_2 e^{\frac{1}{2}t}$,

(c) $x_c = e^{\frac{1}{2}t}(c_1\cos 2t + c_2\sin 2t)$, (d) $y_c = c_2 e^{-2x} + c_2 e^{2x}$,

(e) none of the above. [2]

13. Given the linear second-order ODE

$$4\frac{d^2x}{dt^2} - 4\frac{dx}{dt} + x = 25\sin t,$$

which of the following is a particular integral?

(a) $x_p = -3\cos t + 4\sin t$, (b) $x_p = 4e^t\sin(t + \pi/3),$,

(c) $y_p = -3\cos x + 4\sin x$, (d) $x_p = 4\cos t - 3\sin t$

(e) none of the above. [2]

14. Given the linear second-order ODE

$$4\frac{d^2x}{dt^2} - 4\frac{dx}{dt} + x = 25\sin t\,,$$

satisfying initial conditions $x(0) = 4$ and $x'(0) = -2$, which of the following is the solution?

(a) $y = e^{-2x} - e^{2x} - 3\cos x + 4\sin x$, (b) $x = e^{-\frac{1}{2}t} - e^{\frac{1}{2}t} + 4e^t \sin(t + \pi/3)$,

(c) $x = te^{\frac{1}{2}t} + 4\cos t - 3\sin t$, (d) $x = te^{\frac{1}{2}t} - e^{\frac{1}{2}t} - 3\cos t + 4\sin t$,

(e) none of the above. [2]

15. Let a function of two variables be given as

$$f(x,y) = \tan^{-1}\left(\frac{x}{y}\right)\,.$$

The first- and second-order partial derivatives with respect to x are

(a) $\dfrac{\partial f}{\partial x} = \dfrac{y}{x^2+y^2}$, and $\dfrac{\partial^2 f}{\partial x^2} = \dfrac{2xy}{(x^2+y^2)^2}$,

(b) $\dfrac{\partial f}{\partial x} = \dfrac{x}{x^2+y^2}$, and $\dfrac{\partial^2 f}{\partial x^2} = -\dfrac{2xy}{(x^2+y^2)^2}$,

(c) $\dfrac{\partial f}{\partial x} = -\dfrac{x}{x^2+y^2}$, and $\dfrac{\partial^2 f}{\partial x^2} = \dfrac{2xy}{(x^2+y^2)^2}$,

(d) $\dfrac{\partial f}{\partial x} = \dfrac{y}{x^2+y^2}$, and $\dfrac{\partial^2 f}{\partial x^2} = -\dfrac{2xy}{(x^2+y^2)^2}$,

(e) none of the above. [3]

Section B

Enter your full answers in the answer booklet. This section has three questions, each worth 20 marks.

1. a) Find all complex solutions $z = x + iy$, of the equation $\sin(iz) = \frac{4}{3}$. [7]

b) Obtain the limits [7]

$$(i)\ \lim_{x\to -1} \frac{2-\sqrt{-4x}}{x+1} \quad \text{and} \quad (ii)\ \lim_{x\to\infty} \sqrt[3]{3x}\left[(3x-1)^{2/3} - (3x+1)^{2/3}\right].$$

c) For a function $y(x)$ where both y and x are given in terms of a parameter s, in the form $y = y(s)$ and $x = x(s)$, show that

$$\frac{dy}{dx} = \frac{\frac{dy}{ds}}{\frac{dx}{ds}}$$

and obtain

$$\frac{d^2y}{dx^2}$$

in terms of derivatives of x and y with respect to s. [6]

2. a) Given the function $f(x) = \frac{x^2 - x + 1}{x - 1}$, find all stationary points and use the second derivative to classify them. Find all points of inflection and asymptotes, and sketch the function, explaining your reasoning. You may use results from section A. [6]

b) Obtain the indefinite integral $I = \int \sin^{-1} x \, dx$. Hence, or otherwise, determine the area enclosed by the line $y = \frac{\pi}{2}$, the curve $y = \sin^{-1} x$, and the $y-$axis. [7]

c) Derive the Maclaurin Series for $\cos x$ up to the term in x^4. Use this series to obtain an estimate for $\cos 1$, and use the Lagrange form of the remainder term to give the best bound you can obtain for the error incurred. [7]

3. a) Obtain the solution of the first-order ODE

$$\frac{dy}{dx} = \frac{x+y}{2x-y} + \frac{3y}{2x}$$

satisfying the initial condition $y(1) = 2$. [7]

b) Use Leibnitz' theorem to differentiate the second-order ODE

$$\frac{d^2y}{dx^2} - x\frac{dy}{dx} + xy = 0$$

n times, and let $x = 0$ to obtain the recurrence relation

$$y^{(n+2)}(0) = ny^{(n)}(0) - ny^{(n-1)}(0) \,, \qquad (n \geq 1)$$

where $y^{(k)}(0)$ is $y(x)$ differentiated k times, and evaluated at $x = 0$. Hence, obtain the first five non-zero terms of $y(x)$, the Maclaurin series solution of the ODE, which satisfies the initial conditions $y(0) = 1$ and $y'(0) = -1$. [8]

c) Let a function of two variables be given as

$$f(x,y) = \tan^{-1}\left(\frac{x}{y}\right) \,.$$

Obtain the first- and second-order partial derivatives with respect to y. Is $f(x,y)$ a solution of Laplace's equation, $\nabla^2 f = 0$? You may use results from section A. [5]